

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>

- De Broglie wavelength (λ) = $h / (\sqrt{2 \cdot m \cdot E})$
- m (mass of electron) = 9.11×10^{-31} kg
- E (kinetic energy) = 1 eV = 1.602×10^{-19} J
- h (Planck's constant) = 6.626×10^{-34} Js
- Using the formula , calculate λ .
- Converting units to cm is more appropriate for this question .
- $1 \text{ J} = 1 \text{ kg m}^2/\text{s}^2 = 10^7 \text{ erg}$
- $1 \text{ eV} = 1.602 \times 10^{-12} \text{ erg}$
- Substitute the values and simplify .

</think>

<answer> 2.48×10^{-8} cm</answer>

